HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Managed and monitored **water treatment and waste systems**, ensuring optimal performance and adherence to environmental regulations for critical facility infrastructure.
- Conducted **data analysis and reporting** on system performance metrics, identifying trends to implement process improvements and reduce environmental impact.
- Collaborated with cross-functional teams to support the development and execution of **sustainability initiatives** within Intel Foundry operations.
- Troubleshot and resolved complex **system issues** to minimize downtime, maintaining uninterrupted facility operations and supporting equipment upgrades.
- Supported equipment upgrades, installations, and maintenance activities, ensuring compliance with facility and equipment maintenance practices.
- Developed **Python scripts** for automating data collection and reporting tasks, improving efficiency in monitoring environmental compliance data.
- Performed **production support** for enterprise applications, managing SLAs and responding to P1/P2 incidents in a regulated production environment.

SKILLS

Water/Waste Treatment – Water Treatment Technologies, Waste Treatment Systems, Environmental Regulations, Sustainability Initiatives

Data Analysis & Reporting – Data Analysis Tools, System Performance Monitoring, Trend Identification, Process Improvement

Engineering & Maintenance – Facilities Operations, Equipment Maintenance, System Troubleshooting, Root Cause Analysis

Software & Scripting – Python, SQL, Java, Spring Boot, REST APIs, Docker

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Linux, Git, Jenkins

PROJECTS

Industrial Water Treatment System Optimization

- Designed and implemented a **Python-based data pipeline** to ingest and process real-time sensor data from industrial water treatment systems, capturing operational parameters.
- Developed **SQL queries and scripts** to validate data integrity and extract key performance indicators for system efficiency and environmental compliance reporting.
- Created an **observability dashboard** using Splunk to visualize water treatment system status, alert thresholds, and historical performance trends for operations teams.

Waste Stream Analysis and Reduction Initiative

- Analyzed waste stream data using **pandas and NumPy** to identify major components and sources contributing to environmental impact within a manufacturing context.
- Developed **Python scripts** to automate the aggregation of waste generation data from multiple sources, reducing manual reporting effort by approximately 30%.
- Presented findings and recommendations for waste reduction strategies to the operations and sustainability teams, contributing to process improvement efforts.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)